

BOM G.N.O.S.I.S. Board

| Reference                                                                                                                          | Value                   | Qty | Link                                                                                                                                                                                                                                                                                                                                                                                                         | Footprint                                                                         |
|------------------------------------------------------------------------------------------------------------------------------------|-------------------------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| A1                                                                                                                                 | Arduino_Nano_v2.x       | 1   | <a href="https://www.christians-shop.de/de/mikrocontroller-boards/nano-v3-atmega328p-ch340-usb-c">https://www.christians-shop.de/de/mikrocontroller-boards/nano-v3-atmega328p-ch340-usb-c</a>                                                                                                                                                                                                                | Module:Arduino_Nano_WithMountingHoles                                             |
| C1,C2,C3,C4,C5,C6,C7,C8                                                                                                            | 4.7uF                   | 8   | ~                                                                                                                                                                                                                                                                                                                                                                                                            | Capacitor_THT:CP_Radial_D5.0mm_P2.00mm                                            |
| C9,C10,C11,C12,C13,C14,C15,C16,C17,C19,C20                                                                                         | 100nF                   | 11  | ~                                                                                                                                                                                                                                                                                                                                                                                                            | Capacitor_THT:C_Disc_D4.7mm_W2.5mm_P5.00mm                                        |
| C18,C21,C22                                                                                                                        | 10uF                    | 3   | ~                                                                                                                                                                                                                                                                                                                                                                                                            | Capacitor_THT:CP_Radial_D5.0mm_P2.00mm                                            |
| D1,D2,D3,D4,D5,D6,D7,D8,D9,D10,D11,D12,D13,D14,D15,D16,D17,D18,D19,D20,D21,D22,D23,D24,D25,D26,D27,D28,D29,D30,D31,D32,D33,D34,D35 | 1N4148                  | 35  | <a href="https://www.reichelt.de/de/de/shop/produkt/schalt-diode_100_v_150_ma_do-35-1730">https://www.reichelt.de/de/de/shop/produkt/schalt-diode_100_v_150_ma_do-35-1730</a>                                                                                                                                                                                                                                | Diode_THT:D_DO-35_SOD27_P7.62mm_Horizontal                                        |
| FB1,FB2                                                                                                                            | Ferrit Bead (z. B. 10R) | 2   | <a href="https://www.reichelt.de/de/de/shop/produkt/emv_ferrit-entstoerfilter_axial_90_ohm_100_mhz-105726">https://www.reichelt.de/de/de/shop/produkt/emv_ferrit-entstoerfilter_axial_90_ohm_100_mhz-105726</a>                                                                                                                                                                                              | Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal                     |
| J1, J2, J3, J4, J5, J6 J7, J8 J9, J10, J11, J12, J13, J14, J15, J16, J17, J18, J19                                                 | AudioJack 3,5mm         | 19  | <a href="https://www.3u-shop.de/p/3-5mm-monobuchse-10-stueck">https://www.3u-shop.de/p/3-5mm-monobuchse-10-stueck</a>                                                                                                                                                                                                                                                                                        | AudioJacks:Jack_3.5mm_QingPu_WQP-PJ398SM_Vertical                                 |
| J20                                                                                                                                | POWER Connector         | 1   | <a href="https://www.reichelt.de/de/de/shop/produkt/stiftleiste_2x40-polig_vergoldet_2_54-235639">https://www.reichelt.de/de/de/shop/produkt/stiftleiste_2x40-polig_vergoldet_2_54-235639</a>                                                                                                                                                                                                                | Connector_PinHeader_2.54mm:PinHeader_2x05_P2.54mm_Vertical                        |
| J21                                                                                                                                | Conn_01x10_Socket       | 1   | <a href="https://www.reichelt.de/de/de/shop/produkt/buchsenleiste_2_54mm_1x34_gerade-266681?nbc=1">https://www.reichelt.de/de/de/shop/produkt/buchsenleiste_2_54mm_1x34_gerade-266681?nbc=1</a>                                                                                                                                                                                                              | Connector_PinSocket_2.54mm:PinSocket_1x10_P2.54mm_Vertical                        |
| LED1, LED2, LED3, LED4, LED5, LED6, LED7, LED8, LED9, LED10, LED11, LED12                                                          | LED 3mm FlatTop         | 12  | <a href="https://www.taydaelectronics.com/flat-top-red-led-3mm-diffuse-lens.html">https://www.taydaelectronics.com/flat-top-red-led-3mm-diffuse-lens.html</a>                                                                                                                                                                                                                                                | LED_THT:LED_D3.0mm_FlatTop                                                        |
| Q1,Q2,Q3,Q4,Q5,Q6,Q7,Q8,Q9                                                                                                         | 2N3904                  | 9   | <a href="https://www.reichelt.de/de/de/shop/produkt/bipolartransistor_npn_40v_0_2a_0_5w_to-92-2020">https://www.reichelt.de/de/de/shop/produkt/bipolartransistor_npn_40v_0_2a_0_5w_to-92-2020</a>                                                                                                                                                                                                            | Package_TO_SOT_THT:TO-92_Inline                                                   |
| Q10,Q11                                                                                                                            | BC547                   | 2   | <a href="https://www.reichelt.de/de/de/shop/produkt/bipolartransistor_npn_45v_0_1a_0_5w_to-92-5006">https://www.reichelt.de/de/de/shop/produkt/bipolartransistor_npn_45v_0_1a_0_5w_to-92-5006</a>                                                                                                                                                                                                            | Package_TO_SOT_THT:TO-92_Inline                                                   |
| R1,R2,R3,R4,R6,R7,R8,R9,R10,R12,R14,R16,R18,R20,R22,R24,R26,R27,R28,R29,R30,R31,R32,R33,R34,R36,R38,R39,R40,R41,R42,R43,R44,R68    | 1K                      | 34  | ~                                                                                                                                                                                                                                                                                                                                                                                                            | Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal                     |
| R5                                                                                                                                 | 10M                     | 1   | ~                                                                                                                                                                                                                                                                                                                                                                                                            | Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal                     |
| R11,R13,R15,R17,R19,R21,R23,R25,R35,R37,R47,R50,R53,R56,R59,R66,R67                                                                | 10K                     | 17  | ~                                                                                                                                                                                                                                                                                                                                                                                                            | Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal                     |
| R45,R48,R51,R54,R57                                                                                                                | 100K                    | 5   | ~                                                                                                                                                                                                                                                                                                                                                                                                            | Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal                     |
| R46,R49,R52,R55,R58                                                                                                                | 1M                      | 5   | ~                                                                                                                                                                                                                                                                                                                                                                                                            | Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal                     |
| R60,R61,R64,R65                                                                                                                    | 1,5K                    | 4   | ~                                                                                                                                                                                                                                                                                                                                                                                                            | Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal                     |
| R62, R63                                                                                                                           | 470R                    | 1   | ~                                                                                                                                                                                                                                                                                                                                                                                                            | Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal                     |
| RV11, RV12, RV13, RV14, RV15, RV16, RV17, RV18, RV21, RV22, RV23, RV24, RV25, RV26, RV27, RV28                                     | B100K                   | 16  | <a href="https://www.taydaelectronics.com/100k-ohm-linear-dual-taper-potentiometer-round-shaft-pcb-9mm.html">https://www.taydaelectronics.com/100k-ohm-linear-dual-taper-potentiometer-round-shaft-pcb-9mm.html</a><br><a href="https://www.3u-shop.de/p/potentiometer-metallachse-6-35mm-basis-9mm-100-k-ohm-b-lin">https://www.3u-shop.de/p/potentiometer-metallachse-6-35mm-basis-9mm-100-k-ohm-b-lin</a> | Potentiometer_THT:Potentiometer_Alpha_RD901F-40-00D_Single_Vertical_CircularHoles |
| SW1, SW2, SW3, SW4, SW5, SW6, SW7, SW8                                                                                             | SPDT SubMini            | 8   | <a href="https://www.taydaelectronics.com/sub-mini-toggle-switch-2m-series-spdt-on-off-on-pcb-pins.html">https://www.taydaelectronics.com/sub-mini-toggle-switch-2m-series-spdt-on-off-on-pcb-pins.html</a>                                                                                                                                                                                                  | PCM_4ms_Switch:Switch_Toggle_SPDT_SubMini                                         |
| U1,U2,U3                                                                                                                           | TL074                   | 3   | <a href="https://www.reichelt.de/de/de/shop/produkt/operationsverstaerker_4-fach_dip-14-21557">https://www.reichelt.de/de/de/shop/produkt/operationsverstaerker_4-fach_dip-14-21557</a>                                                                                                                                                                                                                      | Package_DIP:DIP-14_W7.62mm                                                        |
| U4                                                                                                                                 | L78L05_TO92             | 1   | <a href="https://www.reichelt.de/de/de/shop/produkt/spannungsregler_fest_5_v_0_1_a_4_to-92-216616">https://www.reichelt.de/de/de/shop/produkt/spannungsregler_fest_5_v_0_1_a_4_to-92-216616</a>                                                                                                                                                                                                              | Package_TO_SOT_THT:TO-92_Inline                                                   |
| U5                                                                                                                                 | CD4066                  | 1   | <a href="https://www.reichelt.de/de/de/shop/produkt/analog-schalter-ic_4-kanal_dil-14-3234">https://www.reichelt.de/de/de/shop/produkt/analog-schalter-ic_4-kanal_dil-14-3234</a>                                                                                                                                                                                                                            | Package_DIP:DIP-14_W7.62mm                                                        |
| U6                                                                                                                                 | TL072                   | 1   | <a href="https://www.reichelt.de/de/de/shop/produkt/j-fet_operationsverstaerker_2-fach_16_v_s_4_mhz_dip-8-421944">https://www.reichelt.de/de/de/shop/produkt/j-fet_operationsverstaerker_2-fach_16_v_s_4_mhz_dip-8-421944</a>                                                                                                                                                                                | Package_DIP:DIP-8_W7.62mm                                                         |
| U7                                                                                                                                 | MCP4725xxx-xCH          | 1   | <a href="https://www.ebay.de/itm/334629150513">https://www.ebay.de/itm/334629150513</a>                                                                                                                                                                                                                                                                                                                      | Connector_PinHeader_2.54mm:PinHeader_1x05_P2.54mm_Vertical                        |
| U8                                                                                                                                 | MPR121QR2               | 1   | <a href="https://www.amazon.de/Youmile-Capacitive-Ber%C3%BChrungssensor-Entwicklungsboard-Stiftleiste/dp/B09PH3FS2N">https://www.amazon.de/Youmile-Capacitive-Ber%C3%BChrungssensor-Entwicklungsboard-Stiftleiste/dp/B09PH3FS2N</a>                                                                                                                                                                          | Touch_Spirals:Breakout_MPR121                                                     |

BOM G.N.O.S.I.S. Panel

| Reference | Value          | Qty | Link                                                                                                                                                                                                          | Footprint                                                    |
|-----------|----------------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| J1        | Conn_01x10_Pin | 1   | <a href="https://www.reichelt.de/de/de/shop/produkt/stiftleiste_36-polig_gewinkelt_vergold_-235643?nbc=1">https://www.reichelt.de/de/de/shop/produkt/stiftleiste_36-polig_gewinkelt_vergold_-235643?nbc=1</a> | Connector_PinHeader_2.54mm:PinHeader_1x10_P2.54mm_Horizontal |